$$\frac{5}{2}\left(e^{x}-e^{-x}\right)+\frac{1}{2}\left(e^{x}+e^{-x}\right)+5=0$$

$$3e^{x}-2e^{-x}+5=0$$

$$3e^{2x}-2+5e^{x}=0$$

$$e^{x}=\frac{1}{3}$$

$$\ln e^{x}=\ln \tfrac{1}{3}=\ln 3^{-1}=-\ln 3$$

$$1^{15}\left(\cos 15\theta+i\sin 15\theta\right)=-i$$

$$\cos 15\theta=0 \qquad \sin 15\theta=-1$$

$$15\theta=\frac{\pi}{2} \qquad 15\theta=-\frac{\pi}{2},\frac{\pi}{2}$$

$$z^{3}+pz^{2}+6z+q \qquad \alpha,\beta,\gamma$$

$$\alpha^{2}+\beta^{2}+\gamma^{2}=-12$$

$$-12=-p-2(6)$$

$$-12+12=-p$$

$$p=0$$

$$\Sigma\alpha=-p \qquad \Sigma\alpha\beta=6 \qquad \alpha\beta+\beta\gamma+\gamma\alpha=6$$

$$\alpha^{2}+\beta^{2}+\gamma^{2}=12$$

$$\alpha\beta\gamma=-q$$

$$z^{3}-(\alpha+\beta+\gamma)z^{2}+(\alpha\beta+\beta\gamma+\gamma\alpha)z-(\alpha\beta\gamma)$$

$$\lim_{x\to 0}\frac{x+x\tan x}{\sin 3x} = \frac{1+x\sec^2 x+\tan x}{3\cos 3x}$$

$$= 1/3$$

$$\sin^5\theta \equiv \frac{1}{16}\left(10\sin\theta - 5\sin 3\theta + \sin 5\theta\right)$$

$$z+\frac{1}{z} = 2\cos\theta \qquad z^n+\frac{1}{z^n} = 2\cos n\theta$$

$$z-\frac{1}{z} = 2i\sin\theta \qquad z^n-\frac{1}{z^n} = 2i\sin n\theta$$

$$\left(z-\frac{1}{z}\right)^5 = (2i\sin\theta)^5$$

$$32i\sin^5\theta = z^5 + 5z^4(-z^{-1}) + 10(z^3)(-z^{-1})^2$$
$$+10z^2(-z^{-1})^3 + 5z(-z^{-1})^4 + (-z^{-1})^5$$
$$(z^5 - z^{-5}) - 5(z^3 - z^{-3}) + 10(z - z^{-1})$$

$$32i\sin^5\theta = 2i\sin 5\theta - 10i\sin 3\theta + 20i\sin\theta$$